

Practical 2

Data Wrangling II

Create an “Academic performance” dataset of students and perform the following operations using Python.

1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.

[]:

3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and

```
import pandas as pd
import numpy as np
from pathlib import Path
```

```
[10]: df = pd.read_csv('/content/StudentPerformanceFactors.csv')
df
```

```
[10]:
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23	84	Low	High	
1	19	64	Low	Medium	
2	24	98	Medium	Medium	
3	29	89	Low	Medium	
4	19	92	Medium	Medium	
...	...	...	...	...	
6602	25	69	High	Medium	
6603	23	76	High	Medium	
6604	20	90	Medium	Low	
6605	10	86	High	High	
6606	15	67	Medium	Low	

	Extracurricular_Activities	Sleep_Hours	Previous_Scores	\
0	No	7	73	
1	No	8	59	
2	Yes	7	91	
3	Yes	8	98	
4	Yes	6	65	
...	...	...	...	
6602	No	7	76	
6603	No	8	81	

6604	Yes	6	65
6605	Yes	6	91
6606	Yes	9	94

	Motivation_Level	Internet_Access	Tutoring_Sessions	Family_Income	\
0	Low	Yes	0	Low	
1	Low	Yes	2	Medium	
2	Medium	Yes	2	Medium	
3	Medium	Yes	1	Medium	
4	Medium	Yes	3	Medium	
...	...	...	...	...	
6602	Medium	Yes	1	High	
6603	Medium	Yes	3	Low	
6604	Low	Yes	3	Low	
6605	High	Yes	2	Low	
6606	Medium	Yes	0	Medium	

	Teacher_Quality	School_Type	Peer_Influence	Physical_Activity	\
0	Medium	Public	Positive	3	
1	Medium	Public	Negative	4	
2	Medium	Public	Neutral	4	
3	Medium	Public	Negative	4	
4	High	Public	Neutral	4	
...	...	...	...	...	
6602	Medium	Public	Positive	2	
6603	High	Public	Positive	2	
6604	Medium	Public	Negative	2	
6605	Medium	Private	Positive	3	
6606	Medium	Public	Positive	4	

	Learning_Disabilities	Parental_Education_Level	Distance_from_Home	\
0	No	High School	Near	
1	No	College	Moderate	
2	No	Postgraduate	Near	
3	No	High School	Moderate	
4	No	College	Near	
...	...	...	...	
6602	No	High School	Near	
6603	No	High School	Near	
6604	No	Postgraduate	Near	
6605	No	High School	Far	
6606	No	Postgraduate	Near	

	Gender	Exam_Score
0	Male	67
1	Female	61
2	Male	74

3	Male	71
4	Female	70
...	...	...
6602	Female	68
6603	Female	69
6604	Female	68
6605	Female	68
6606	Male	64

[6607 rows x 20 columns]

```
[11]: # 1. SCAN & DEAL WITH MISSING VALUES AND INCONSISTENCIES
print("--- Step 1: Missing Values ---")
print(df.isnull().sum())    # Scan for nulls

# Handle Inconsistencies: standardizing categorical text (strip spaces & fix_
# casing)
for col in df.select_dtypes(include='object').columns:
    df[col] = df[col].str.strip().str.title()
print("\nInconsistencies handled (Text standardized to Title Case).")
```

```
--- Step 1: Missing Values ---
Hours_Studied      0
Attendance          0
Parental_Involvement  0
Access_to_Resources  0
Extracurricular_Activities  0
Sleep_Hours        0
Previous_Scores     0
Motivation_Level    0
Internet_Access     0
Tutoring_Sessions   0
Family_Income       0
Teacher_Quality     78
School_Type         0
Peer_Influence      0
Physical_Activity    0
Learning_Disabilities  0
Parental_Education_Level  90
Distance_from_Home   67
Gender              0
Exam_Score          0
dtype: int64
```

Inconsistencies handled (Text standardized to Title Case).

```
[12]: # 2. SCAN & DEAL WITH OUTLIERS
print("\n--- Step 2: Outliers (IQR Method) ---")
numeric_cols = df.select_dtypes(include=[np.number]).columns

for col in numeric_cols:
    Q1, Q3 = df[col].quantile(0.25), df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower, upper = Q1 - 1.5 * IQR, Q3 + 1.5 * IQR

    # Identify count
    outlier_count = ((df[col] < lower) | (df[col] > upper)).sum()
    print(f"{col}: {outlier_count} outliers found.")

    # Handle: Capping (Winsorization)
    df[col] = np.clip(df[col], lower, upper)
```

```
--- Step 2: Outliers (IQR Method) ---
Hours_Studied: 43 outliers found.
Attendance: 0 outliers found.
Sleep_Hours: 0 outliers found.
Previous_Scores: 0 outliers found.
Tutoring_Sessions: 430 outliers found.
Physical_Activity: 0 outliers found.
Exam_Score: 104 outliers found.
```

```
[15]: # --- Step 3: Data Transformation ---
# 1. Log Transformation to decrease skewness
df['Hours_Studied_log'] = np.log1p(df['Hours_Studied'])

# 2. Z-Score Scaling to change the scale (standardization)
df['Exam_Score_scaled'] = (df['Exam_Score'] - df['Exam_Score'].mean()) / \
    df['Exam_Score'].std()

print("--- Step 3: Transformation Results ---")
print(df[['Hours_Studied', 'Hours_Studied_log', 'Exam_Score', \
    'Exam_Score_scaled']].head())
```

```
--- Step 3: Transformation Results ---
```

	Hours_Studied	Hours_Studied_log	Exam_Score	Exam_Score_scaled
0	23	3.178054	67	-0.037694
1	19	2.995732	61	-1.816574
2	24	3.218876	74	2.037667
3	29	3.401197	71	1.148227
4	19	2.995732	70	0.851746

```
[ ]:
```